Competing-Risks Modelling of a Widely-Used Cirrhosis Outcome Benchmark: Recovering the Transplant Outcome and Cause-Specific Risk Lost to Classification

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Abstract

The Mayo primary biliary cirrhosis (PBC) cohort (418 patients), a widely-used machine-learning benchmark, is right-censored with *competing* death and transplantation, and is widely analysed as classification of its end-of-follow-up status label (alive, transplanted, deceased). We recast it as competing-risks survival against the five-variable Mayo score. In a simulation fitted to the cohort (cause-specific Weibull hazards; n = 418–4000; 200 replicates), random forests ranked true 5-year transplant risk at Spearman 0.56 (0.59 with SMOTE) at n = 418, against 0.91 for multinomial logistic regression, 0.92 for cause-specific Cox and 0.90 for Fine–Gray. Transplant F1 stayed at 0.14–0.25 for every classifier, so F1 does not measure transplant-risk recovery. Our real-data tree-ensemble classifiers reached transplant-class F1 0.007–0.180. Under leakage-free cross-validation, seven survival models, deep networks included, reached Harrell's C 0.81–0.83. None exceeded the Mayo score (C = 0.833, integrated Brier 0.143), so the static variables set the ceiling. As a competing event, transplant reached C 0.82 (25 events; 95% CI 0.80–0.84). Age acted in opposite directions (cause-specific HR 1.36 for death, 0.74 for transplant per SD; subdistribution HR 1.52, 0.32). In competing-risks decision curves for 5-year death, Cox and Fine–Gray matched the Mayo score's net benefit (largest differences 0.0145, 0.0172), and every model exceeded treat-all and treat-none. Longitudinally, baseline death-risk concordance fell from 0.87 to 0.64 over six years and was restored to 0.78 by the latest measurement (paired ΔC up to +0.14). For time-resolved absolute risk of competing events on small tabular clinical data, the competing-risks formulation and risk-based evaluation matter more than model sophistication.

Keywords: liver cirrhosis; survival analysis; competing risks; machine learning; benchmark re-examination; reproducibility; clinical risk prediction

1 Introduction

A recurring methodological problem in clinical prediction is that right-censored time-to-event data with competing events are analysed as multi-class classification. Such a target cannot give the time-resolved absolute risk of each competing outcome that prognosis requires. It discards follow-up time and treats a competing event as a static outcome label, biasing absolute-risk estimates and obscuring cause-specific structure. The consequences are concrete and measurable in a widely-used cirrhosis outcome benchmark, which we use as the worked instance throughout this paper.

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Liver cirrhosis is a leading cause of morbidity and mortality worldwide, responsible for about one million deaths annually and a substantial healthcare burden [1]. Accurate prognostication is central to its management. It guides the timing of interventions, the prioritisation of candidates for liver transplantation, and the allocation of scarce resources. Decades of clinical research have produced validated prognostic scores for this purpose, of which the Mayo PBC natural-history model [2] and the Model for End-Stage Liver Disease (MELD) [3] are the most established.

The public "Cirrhosis Patient Survival Prediction" dataset [2, 4], comprising 418 patients from the Mayo Clinic PBC trial, has become a popular machine-learning (ML) benchmark. It is commonly analysed as a *classification* problem. A multi-class prediction competition was built on the recorded status label (alive, transplanted, or deceased) [5], applied studies classify the data with tree ensembles and neural networks [6], and Tabatabai et al. classified death status and scored the classifiers by F-score and accuracy [7]. The standard remedy for class imbalance, such as the rarity of transplant, is synthetic minority oversampling (SMOTE) [8]. Under the same covariates and cross-validation as our survival models, four classifiers of the status label (random forest and gradient boosting, each with and without SMOTE) reached transplant-class F1 0.007 to 0.180, macro-F1 0.505 to 0.542 and accuracy 0.693 to 0.742 (Section 3.6).

This framing answers the wrong question for prognosis. The dataset records, for each patient, a follow-up time and a terminal status. Patients still alive at last contact are *right-censored*: their event time is unknown beyond that contact. A patient who receives a transplant is removed from the at-risk pool for death. Death and transplantation are therefore *competing risks* [9, 10]. Predicting the realised status under fixed administrative follow-up is itself a coherent task, but a status label scored by per-class F1 discards the follow-up time and gives no time-resolved absolute risk of either competing event. The underlying arguments are established in general. Treating censored cases as non-events biases prediction, and inverse probability of censoring weighting is the remedy [11]. Classification of long-horizon outcomes is biased by censoring [12], survival outcomes are commonly binarised for classification [13], and resampling such as SMOTE miscalibrates risk models [14]. We apply these arguments to this benchmark. A low transplant-class F1 does not by itself show that transplant risk is lost: in our simulation (Section 4.8), multinomial logistic regression reaches transplant F1 0.15 with a near-true risk ranking (Spearman 0.91), whereas random forests lose the ranking (0.56). For the clinically relevant absolute-risk and cause-specific estimands, the appropriate formulation is time-to-event survival analysis with competing risks.

This paper re-examines that framing. We quantify what the classification framing costs for risk prediction, apply the competing-risks formulation, and release a reproducible pipeline so the benchmark can be analysed with it. Our contributions are:

1. We fit a simulation to the cohort (cause-specific Weibull hazards, resampled covariates, Kaplan–Meier censoring, $n = 418$, 1000 and 4000) and compare classifiers scored by F1 with competing-risks models on recovery of the true 5-year transplant risk. Random forests recover the ranking poorly (Spearman 0.55 to 0.61 across n), cause-specific Cox, Fine–Gray and multinomial logistic regression recover it (0.90 to 0.99), and transplant F1 stays at 0.14 to 0.25 whatever the ranking. The closest designs are a PBC-calibrated single-event simulation that compares Cox with random survival forests, without classifiers or F1 [15], and the `simPBC` generator of `riskRegression`, which simulates PBC-like data and is not used there to compare classifiers with competing-risks models [16]. To our knowledge, no prior work has compared classifier and competing-risks formulations on recovery of true cause-specific risk.
2. We apply established arguments against scoring censored outcomes as classification labels [11–14] to this benchmark, with a classification baseline under the same cross-validation as the survival models (transplant-class F1 0.007 to 0.180).

3. We evaluate established competing-risks models (cause-specific Cox, Fine–Gray, two-event DeepHit), benchmarked on these data in prior work [17–20], against the Mayo risk score [2], including competing-risks decision-curve analysis (DCA) for death at 5 years with the Mayo score as a strategy. The Mayo-anchored comparison is the new part.
4. We benchmark seven survival models, from the historical Mayo clinical score to deep neural survival networks (DeepSurv, DeepHit), under a leakage-free protocol with survival-appropriate metrics. Discrimination is *flat* (Harrell's C 0.81 to 0.83) and no model exceeds the five-variable Mayo score, so the information in the static feature set caps discrimination at every level of model capacity tested, consistent with the broader finding that simple models are hard to beat on small tabular clinical data.
5. We model transplantation as a competing event, discriminated with cross-validated C = 0.82 (25 events; 95% CI 0.80–0.84). The cause-specific and subdistribution analyses give the age effect on each event (cause-specific HR 1.36 for death and 0.74 for transplant per SD; subdistribution HR 1.52 and 0.32), which a status label scored by F1 does not report. The opposite directions are consistent with referral and listing practice in PBC, where wait-list mortality among listed patients is high [21].
6. We extend the analysis to the *longitudinal* cohort, showing with time-varying competing-risks models and landmarking that a baseline measurement's prognostic value decays over follow-up and is recovered by updating biomarkers.
7. We show, across three further clinical datasets, that the benefit of model complexity scales with sample size and is negligible at the cirrhosis cohort's scale, which places the 418-patient cohort in the range where formulation matters more than model complexity.
8. We release the complete pipeline, which is free of leakage, as an openly reproducible framework for the analysis of competing risks in this and comparable clinical cohorts.

2 Background and Related Work

Clinical prognostic scores. Cirrhosis prognosis has long relied on rule-based scores. The Child–Pugh score and MELD [3] are widely used, the latter combining serum creatinine, bilirubin and the international normalised ratio (INR) to predict short-term mortality. For primary biliary cirrhosis specifically, the Mayo natural-history model [2] predicts survival from five routinely available variables (bilirubin, albumin, age, prothrombin time and edema) and was derived on the very cohort distributed as this dataset. MELD cannot be computed here, because the dataset contains neither serum creatinine nor INR. Its `Prothrombin` field is prothrombin time in seconds, from which INR is not recoverable without the laboratory ISI and control time. The Mayo PBC score, by contrast, uses only variables present in the data and is therefore the appropriate clinical baseline. Two contemporary PBC-specific scores, the GLOBE score [22] and the UK-PBC risk score [23], are likewise non-computable here, as they require treatment-response biochemistry and markers absent from this dataset. More recent work has explored dynamic, longitudinally-updated models for cirrhosis survival [24].

ML on this dataset. ML studies on this dataset commonly treat it as classification of the status label. Representative examples classify the data with tree ensembles and neural networks [6] or with new regression-based classifiers scored by F-score and accuracy [7], and the dataset underpinned a multi-class prediction competition [5]. Such studies use `N_Days` only as a static feature, if at all, and treat transplantation as an ordinary label. Obite et al. [25] compared survival models

on this cohort, a proposed factor-enhanced DeepSurv with Cox, random survival forests, DeepSurv and DeepHit, by concordance and Brier score. Competing-risks analyses of these data also exist. Ishwaran et al. fitted competing-risks random survival forests to the `pbc` data [17], and Chen fitted competing-risks DeepHit to PBC [18]. Zhuang et al. fitted cause-specific Cox, random survival forest, Fine–Gray and cause-specific Weibull models to the 312 PBC trial patients with the Mayo-score covariates [19]. Sierra et al. benchmarked cause-specific Cox, Fine–Gray, penalised Fine–Gray, random survival forests, DeSurv and DeepHit on the 418-patient `pbc` data, with decision-curve analysis at 5 years [20]. None of these competing-risks analyses compares the models with the Mayo score. This paper adds that comparison and the simulation.

Survival machine learning. Beyond the semiparametric Cox proportional hazards model [26], ML survival methods include random survival forests [27] and, more recently, neural approaches such as DeepSurv (a neural Cox model) [28] and DeepHit (a discrete-time model with native competing-risks support) [29]. Whether such models outperform Cox or simple clinical scores on small tabular datasets is contested. On general tabular data, a recurring finding is that tree ensembles and well-specified simple models are hard to beat with deep networks when sample sizes are limited [30, 31]; we test whether this pattern holds in the survival setting.

Competing risks. When several event types preclude one another, analysing one while censoring the others biases absolute-risk estimates [9]: Kaplan–Meier estimation with the competing event censored overestimates cumulative incidence, by a pooled ratio of 1.41 across studies [32]. Two complementary models are standard, each answering a different question [33]. Cause-specific Cox models (one hazard per event type) address aetiological questions, while the Fine–Gray subdistribution-hazard model [10] addresses absolute risk, with the Aalen–Johansen estimator [34] providing nonparametric cumulative incidence. A cause-specific hazard ratio describes a covariate's effect on the event rate among patients still event-free, and a subdistribution hazard ratio its effect on the cumulative incidence, so the two can differ in size and in direction for the same covariate [33, 35].

Classification and time-to-event formulations. Logistic regression and proportional-hazards models give similar estimates when the outcome is rare and follow-up short [36, 37], and when the disease is rare, the risk factor moderate and censoring rates similar across covariates [38]. Survival stacking recasts a survival regression as classification on a person-period data set and, with logistic regression, approximates the Cox model [39]. Binary classification under right-censoring has been studied without competing risks [40], and the bias from treating competing events as censoring has been characterised theoretically [41]. Statistical and machine-learning models for competing risks have been compared [42, 43], and none of the 10 simulation studies in a scoping review of statistical and machine-learning time-to-event comparisons was calibrated to PBC [44]. The closest PBC-based designs are a PBC-calibrated single-event simulation comparing Cox with random survival forests [15] and the `simPBC` generator of `riskRegression` [16]. None compares multiclass classification with competing-risks models on recovery of true cause-specific risk.

3 Materials and Methods

3.1 Dataset and cohort

We used the Mayo PBC cohort (UCI "Cirrhosis Patient Survival Prediction") [2, 4]. The cohort is specifically primary biliary cirrhosis (now primary biliary cholangitis), an autoimmune cholestatic disease, so it is not representative of cirrhosis from viral, alcoholic or metabolic causes; we retain the historical name to match the dataset. It comprises 418 patients with a follow-up time (`N_Days`) and terminal status ("C" for censored/alive, $n = 232$; "D" for death, $n = 161$; "CL" for censored at liver transplant, $n = 25$), plus 17 demographic, clinical and laboratory variables. Median follow-up

was 1730 days (range 41–4795). Several laboratory variables had substantial missingness, reflecting the well-known structure in which 106 patients enrolled without the full panel. Missing values were imputed inside each cross-validation fold (below).

3.2 Survival reformulation

We replaced the three-class target with time-to-event structure. For overall survival, the event was death and transplantation was treated as a competing event (cause-specific censoring). We additionally defined the transplant endpoint, with transplant as the event and death censored. For the competing-risks description we used the event code $\{0 : \text{censored}, 1 : \text{death}, 2 : \text{transplant}\}$.

3.3 Clinical baseline: the Mayo PBC score

As the clinical reference we computed the Mayo natural-history risk score [2],

$$\begin{aligned} R = {} & 0.871 \ln(\text{bilirubin}) - 2.53 \ln(\text{albumin}) \\ & + 0.039\,\text{age} + 2.38 \ln(\text{prothrombin}) + 0.859\,\text{edema}, \end{aligned}$$

with edema coded 0/0.5/1. We used the score as a single-covariate Cox model, fitting the baseline hazard within each training fold so that it produces calibrated absolute-risk curves comparable to the other models.

3.4 Survival models

We compared seven models: (i) the Mayo score; (ii) Cox proportional hazards [26] with ridge stabilisation; (iii) penalized Cox (elastic net); (iv) random survival forests [27]; (v) gradient-boosted survival (Cox loss); (vi) DeepSurv [28]; and (vii) DeepHit [29]. All models except the Mayo score used the 17 variables. The cross-validated endpoint models (Table 4) and the four-predictor Fine–Gray and two-event DeepHit models (Table 5) used a parsimonious set of age, bilirubin, albumin and prothrombin time. Hyperparameters were fixed in advance, without a search. Cox used a ridge penalty of 1.0 and the elastic net a mixing parameter of 0.5. The random survival forest used 300 trees with a minimum leaf size of 15, and gradient boosting used 300 stages with learning rate 0.05, depth 2 and subsampling fraction 0.8. DeepSurv and DeepHit used two hidden layers of 32 units with dropout 0.3, batch size 64 and at most 256 epochs.

3.5 Preprocessing

All preprocessing was fit *inside* each training fold to prevent leakage, which arises when imputation or scaling is fit on the full dataset before splitting. Numerical variables were median-imputed, log-transformed where skewed, and standardised. Categorical variables were mode-imputed and one-hot encoded. No survival model was trained on oversampled data, because resampling distorts the censoring and event-time distribution and miscalibrates risk [14]. SMOTE appears only in two arms of the classification baseline (Section 3.6), which test the standard remedy for class imbalance, and is fit inside each training fold.

3.6 Classification baseline

To measure the classification framing on the same data, we fitted four three-class classifiers to the status at end of follow-up (C, CL, D), with follow-up time discarded: a random forest (300 trees), the same with SMOTE [8] (5 neighbours) to test the standard remedy for class imbalance, gradient

boosting (300 stages, learning rate 0.05, depth 2, subsampling fraction 0.8), and the same with SMOTE. Features were the 17 baseline covariates of the seven-model comparison through the same preprocessor, with follow-up time and status excluded. Preprocessing and SMOTE were fit on the training fold only, inside an `imbalanced-learn` pipeline, so oversampling never touched the test fold. Evaluation used repeated stratified 5-fold cross-validation (10 repeats, stratified on the label, seed 42) and reports per-class F1, macro-F1 and accuracy as the mean over the 50 folds with a 95% percentile bootstrap interval of the mean.

3.7 Competing-risks analysis

We estimated cause-specific cumulative incidence functions for death and transplantation with the Aalen–Johansen estimator [34], and fit cause-specific Cox models for each event (all covariates for death; a parsimonious three-covariate model of age, log bilirubin and albumin for the rare transplant endpoint). Continuous covariates were standardised, so hazard ratios are per standard deviation.

We also fitted the Fine–Gray subdistribution-hazard model [10] for death and for transplant on the same covariate sets, by the weighted-data estimator of R `survival::finegray` [45], in which a patient who had the competing event stays in the risk set with an inverse-probability-of-censoring weight, and with a subject-clustered robust variance. Its coefficients agreed with those of R `survival::finegray` with `coxph` on the same data within 0.0001, and with R `cmprsk::crr` within 0.005. Both model types are established on PBC [19, 20], and we interpret cause-specific and subdistribution hazard ratios following Wolbers et al. and Austin et al. [33, 35]. For cross-validated discrimination, Fine–Gray models on the four endpoint predictors were scored on the predicted 5-year cumulative incidence with the truncated inverse-probability-of-censoring-weighted concordance for competing risks [46], under the cross-validation of the endpoint models, with Harrell's C (competing event censored) for comparison. Calibration compared the mean out-of-fold predicted 5-year cumulative incidence, averaged over the 10 cross-validation repeats, with the Aalen–Johansen incidence by predicted-risk quintile, with 95% intervals from 1000 bootstrap resamples within each quintile.

3.8 Two-event DeepHit

To test a deep competing-risks model on the same metric, we fitted the two-cause DeepHit [29] (`pycox` `DeepHit`), one network whose output of 2×40 time cuts is read as one probability mass per cause. Architecture and training followed the single-event DeepHit: two hidden layers of 32 units with batch normalisation and dropout 0.3, Adam with learning rate 0.01, $\alpha = 0.2$ and $\sigma = 0.1$, batch size 64, at most 256 epochs, and early stopping with patience 15 on a 20% inner validation split of the training fold. Cross-validation followed the endpoint models, and two covariate sets were run: the four endpoint predictors and the full preprocessed covariates. Both were scored with the 5-year competing-risks concordance used for Fine–Gray and with Harrell's C with the competing event censored.

3.9 Decision-curve analysis

We assessed clinical utility for death at 5 years by competing-risks decision-curve analysis [47], comparing the Mayo PBC score, Cox PH, random survival forests, gradient boosting and a Fine–Gray model for death with treat-all and treat-none. Each strategy's risk is the out-of-fold predicted 5-year cumulative incidence of death with transplant as a competing event, so each threshold is a threshold on the absolute 5-year risk of death. For the Mayo score, Cox PH, random survival forests and gradient boosting, the same model family, with the same settings and in-fold preprocessing,

was fit twice inside each training fold (death with transplant censored, and transplant with death censored; for the Mayo score both are the one-covariate Cox model), and the two predicted cumulative hazards were combined into the cause-specific cumulative incidence of death on the training fold's event times up to 5 years. The Fine–Gray model for death (same in-fold covariates, ridge penalty 1.0) gives that incidence directly. Net benefit at threshold p_t is

$$\mathrm{NB}(p_t) = P(r \geq p_t)\, F_{\mathrm{high}} - P(r \geq p_t)\,(1 - F_{\mathrm{high}})\,\frac{p_t}{1 - p_t},$$

where F_{high} is the Aalen–Johansen 5-year death incidence among patients with predicted risk $r \geq p_t$, so transplant enters as a competing event. Thresholds ran from 0.05 to 0.50 in steps of 0.01. Out-of-fold risks came from the 4-repeat cross-validation of the seven-model comparison, net benefit was computed per repeat and averaged over the repeats, and 95% intervals came from 1000 patient bootstrap resamples. Transplant was not analysed: its 5-year cumulative incidence (4.5%) lies below the lowest threshold, and it is an allocation outcome, decided by listing and organ supply, with no intervention whose harm-benefit trade-off a risk threshold expresses.

3.10 Dynamic analysis on the longitudinal cohort

The static benchmark uses a single measurement per patient. To test whether updating biomarkers over follow-up improves prediction, we additionally analysed the sequential Mayo PBC data (`pbcseq`: 312 patients, 1945 serial measurements), which carries the same competing outcomes. We fit time-varying cause-specific Cox models (counting-process format) for death and transplantation using the most recent bilirubin, albumin, prothrombin time, edema and age. We then performed a landmark analysis at 2, 4 and 6 years. Among patients still at risk, we predicted 3-year death risk from either the baseline (day-0) measurement or the most recent measurement carried forward, comparing concordance between the two covariate sets on identical repeated stratified 5-fold splits.

3.11 Generalisation across datasets

To test whether the limited benefit of model complexity is specific to the small cirrhosis cohort or reflects a broader pattern, we replicated the simple-versus-complex comparison on three widely-used public clinical survival datasets via `pycox`: METABRIC ($n = 1904$), GBSG ($n = 2232$) and SUPPORT ($n = 8873$). On each we compared Cox against random survival forests, gradient boosting and DeepSurv under repeated stratified 5-fold cross-validation with standardisation fit inside each training fold and 200-tree ensembles, reporting Harrell's concordance.

3.12 Transportability to an independent cohort

We assessed whether a model fit on the Mayo PBC cohort transports to an independent cirrhosis cohort, using the Cirrhosis subgroup of the SUPPORT study ($n = 508$). Only markers common to both cohorts were available, namely bilirubin, albumin and age, because SUPPORT records neither prothrombin time nor edema. We therefore trained a 3-covariate cause-specific death model on the PBC cohort and applied it to SUPPORT. SUPPORT enrolled acutely hospitalised, severely ill patients, so case-mix and baseline risk differ markedly between the two cohorts. We accordingly evaluated *discrimination* (transportability of risk ordering, Harrell's C) rather than absolute-risk calibration, for the death endpoint, since transplant is not ascertained as a competing event in SUPPORT.

3.13 Simulation study

To test which formulations recover transplant risk when the truth is known, we simulated data from a mechanism fitted once to the cohort and then fixed. Covariates were the five Mayo variables (age in years, log bilirubin, albumin, log prothrombin time, edema 0/0.5/1), drawn as real baseline vectors resampled with replacement from the 416 complete cases. Event times came from cause-specific Weibull proportional-hazards models for death and transplant, $H_k(t \mid x) = \exp(b_k + x^\top \beta_k)\, t^{\rho_k}$, each fitted by maximum likelihood to the cohort with the other event censored. The fitted age coefficient is $+0.039$ per year for death and -0.091 per year for transplant, with shapes $\rho = 1.49$ and 1.92. Censoring was independent, drawn from the Kaplan–Meier estimate of the real censoring distribution. The truth was each patient's 5-year cumulative incidence of each cause, computed by Gauss-Legendre quadrature (maximum absolute error 1.5×10^{-9}). The simulated event proportions match the cohort (death 37.8% vs. 38.5%, transplant 6.6% vs. 6.0% at $n = 418$). Training sizes were $n = 418$, 1000 and 4000, with 200 replicates each, and every replicate was scored on one fixed simulated test set of 5000 patients. Five approaches used the same five covariates: (a) a three-class random forest, with transplant risk $P(\mathrm{CL})$; (b) the same with SMOTE; (c) unpenalised multinomial logistic regression, $P(\mathrm{CL})$; (d) cause-specific Cox models for both causes, with the 5-year cumulative incidence as risk; and (e) Fine–Gray models for both causes, likewise. Metrics were the Spearman correlation and Harrell's C between each risk score and the true 5-year cumulative incidence, for transplant and for death; the transplant-class F1 of (a) to (c) against status at end of follow-up; and the sign and size of the estimated age effect. The target of the cause-specific Cox age coefficient is the mechanism's coefficient, and the targets for (c) and (e) are their large-sample (least-false) values under the mechanism (multinomial -0.092 for transplant and 0.043 for death, Fine–Gray -0.105 and 0.046, per year). Summaries are the mean and the 2.5 and 97.5 percentiles over replicates. The mechanism has proportional cause-specific hazards that are linear in the covariates. The `simPBC` generator of `riskRegression` also simulates PBC-like competing-risks data [16], and Graf et al. calibrated a single-event simulation to the PBC trial to compare Cox with random survival forests [15]. Neither compares classifiers scored by F1 with competing-risks models.

3.14 Evaluation

Models were evaluated by repeated stratified 5-fold cross-validation, with the held-out fold of each split as the test set. The seven-model comparison, the score augmentation and the paired and refit checks against the Mayo score used 4 repeats, stratified on event type to balance the rare transplant cases, and the decision-curve analysis used the out-of-fold risks of this protocol. The endpoint-specific models (Table 4), the Fine–Gray models and two-event DeepHit used 10 repeats with the same stratification, the classification baseline 10 repeats stratified on the status label, and the internal concordance of the transportability model 10 repeats stratified on death. The landmark analysis used 20 repeats and the three external datasets 2 repeats, each stratified on its event indicator. The simulation used 200 replicates per training size, scored on a fixed test set of 5000 patients. The out-of-fold predictions behind the Cox calibration and the risk-group analyses came from a single stratified 5-fold split, and the Fine–Gray calibration used out-of-fold predictions averaged over the 10 repeats. Within each training fold, DeepSurv and DeepHit held out a random 20% of patients as a validation set for early stopping (patience 15 epochs). The test fold was never used for fitting, preprocessing or early stopping. Confidence intervals were obtained by bootstrapping the per-fold metric values (2000 resamples). They reflect cross-validation variability and omit independent patient-level resampling, so on the rarest strata (the 25-event transplant

Table 1: Cohort characteristics by outcome. Continuous values are median [IQR].

	Overall	Alive	Death	Transpl.
n	418	232	161	25
Age, y	51 [43–58]	50 [41–57]	53.5 [47–61]	41 [37–47]
Bilirubin, mg/dL	1.4 [0.8–3.4]	0.9 [0.6–1.6]	3.2 [1.4–7.1]	3.1 [1.3–3.5]
Albumin, g/dL	3.5 [3.2–3.8]	3.6 [3.4–3.8]	3.4 [3.1–3.6]	3.5 [3.2–3.8]
Prothrombin, s	10.6 [10–11.1]	10.4 [9.9–10.8]	11.0 [10.5–11.8]	10.3 [10–10.7]
Mayo score	4.8 [4.1–5.9]	4.3 [3.8–4.8]	6.0 [5.0–6.8]	5.0 [4.4–5.2]
Ascites	6%	<1%	14%	0%
Stage 4	34%	22%	52%	40%

endpoint and the later landmarks) they should be read as lower bounds on the true sampling uncertainty. The metrics were Harrell's [48] and Uno's [49] concordance indices, time-dependent AUC at 1/3/5 years, and the integrated Brier score [50]. Competing-risks models were also scored with the truncated inverse-probability-of-censoring-weighted concordance for competing risks at 5 years [46]. Calibration was assessed at five years for the Cox model against Kaplan–Meier estimates, and for Fine–Gray against the Aalen–Johansen cumulative incidence by predicted-risk quintile. Reporting follows the TRIPOD guideline [51].

3.15 Implementation

Analyses used `scikit-survival` 0.27 [52], `lifelines` 0.30, `pycox` 0.3 and `imbalanced-learn` 0.14.2 (for SMOTE) in Python 3.13, with a fixed random seed (42) throughout for reproducibility. Every cross-validation fold of the deep models seeds `torch`, `numpy` and Python's `random` from 42 plus the fold index and uses deterministic `torch` kernels on CPU, so a rerun gives identical results. The Fine–Gray estimator was implemented in Python and cross-checked against R `cmprsk::crr` and `survival::finegray` with `coxph` (Section 3.7). The proportional-hazards assumption was checked by Schoenfeld residual tests. Code and the full pipeline are available (see Data Availability).

4 Results

4.1 Cohort

Table 1 summarises the cohort by outcome. Deceased patients were older, with higher bilirubin and Mayo scores. Transplanted patients were markedly *younger* (median age 41 vs. 53.5 years for deaths) at a similar median bilirubin (3.1 vs. 3.2 mg/dL, though over a narrower range), an early signal of the competing-risks structure.

4.2 Cumulative incidence of competing outcomes

The Aalen–Johansen cumulative incidence (Fig. 1) shows death accumulating steadily (7.2%, 19.7% and 29.2% at 1, 3 and 5 years) while transplantation remained rare (0%, 2.4%, 4.5%). Treating transplantation as just another class ignores both its rarity and its competing relationship with death.

4.3 Discrimination is flat across model families

All seven models achieved high discrimination within a narrow range for the death endpoint, and none was significantly better than the Mayo score (Table 2; Supplementary Fig. S1). Harrell's

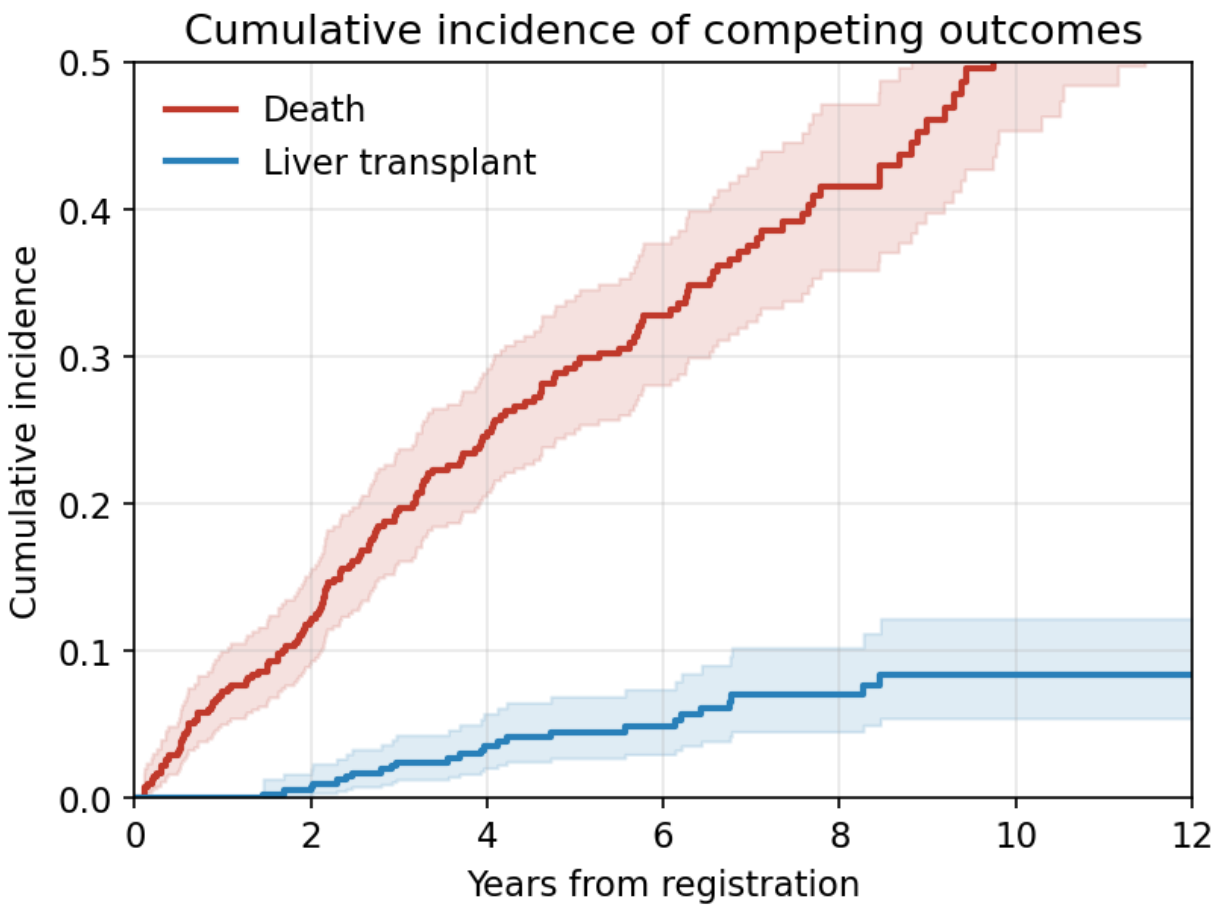


Figure 1: Cumulative incidence of death and liver transplantation (Aalen–Johansen, with 95% confidence bands).

C ranged only from 0.806 (DeepHit) to 0.833 (Mayo). No ML model, whether linear, penalized, ensemble or deep, outperformed the five-variable Mayo clinical score, which also achieved the best integrated Brier score (0.143). The deep models (DeepSurv, DeepHit) were no better, and DeepHit calibrated worst (IBS 0.201), consistent with deep discrete-time models being data-hungry. This flatness indicates that the ∼0.83 ceiling is set by the information in this static set of variables; the longitudinal analysis below shows that further signal becomes available when measurements are updated over time.

Two checks support this reading. First, paired per-fold comparisons against the Mayo score test the flatness statistically. Cox (ΔC −0.003 [−0.011, +0.004]) and penalized Cox (−0.001 [−0.008, +0.006]) were indistinguishable from the score. Random survival forests (−0.009 [−0.018, −0.0004]), gradient boosting (−0.013 [−0.023, −0.005]), DeepSurv (−0.014 [−0.024, −0.006]) and DeepHit (−0.027 [−0.038, −0.017]) ranked slightly below it, with paired intervals below zero. Second, the ceiling is not an artefact of the Mayo score's coefficients having been derived on this cohort: refitting a Cox model on the same five variables *inside* each fold gave matching discrimination (C = 0.832 vs. 0.833 for the fixed score; paired ΔC −0.001 [−0.006, +0.004]), so the externally-fixed score holds no in-sample advantage here.

Table 2: Death-endpoint discrimination and calibration across model families (repeated stratified CV; 95% bootstrap CI). Best value per column in **bold**.

Model	**Harrell's C**	**Uno's C**	**AUC @ 5y**	**IBS** (lower better)
Mayo PBC score	**0.833** [0.815, 0.852]	**0.787** [0.764, 0.809]	**0.907** [0.886, 0.926]	**0.143** [0.130, 0.157]
Cox PH	0.830 [0.813, 0.847]	0.774 [0.754, 0.791]	0.904 [0.883, 0.922]	0.151 [0.138, 0.165]
Coxnet (elastic net)	0.832 [0.815, 0.849]	0.778 [0.760, 0.796]	0.906 [0.884, 0.924]	0.150 [0.136, 0.163]
Random survival forest	0.824 [0.808, 0.840]	**0.787** [0.770, 0.805]	0.892 [0.873, 0.910]	0.152 [0.143, 0.161]
Gradient boosting	0.820 [0.804, 0.835]	0.772 [0.758, 0.786]	0.891 [0.871, 0.908]	0.152 [0.140, 0.165]
DeepSurv	0.819 [0.802, 0.836]	0.774 [0.755, 0.794]	0.891 [0.868, 0.911]	0.159 [0.146, 0.172]
DeepHit	0.806 [0.789, 0.823]	0.769 [0.752, 0.786]	0.882 [0.861, 0.902]	0.201 [0.182, 0.221]

Table 3: Harrell's C across three external clinical survival datasets (repeated stratified CV).

Dataset	n	**Cox**	**RSF**	**GBM**	**DeepSurv**
METABRIC	1904	0.635	0.643	0.642	0.614
GBSG	2232	0.661	0.673	0.675	0.672
SUPPORT	8873	0.570	0.615	0.615	0.611

4.4 Augmenting the clinical score does not help

Adding the biomarkers the Mayo score omits (copper, SGOT, platelets, cholesterol, alkaline phosphatase, fibrosis stage, ascites, and others) on top of the score yielded no improvement. The paired ΔHarrell's C was +0.0008 (95% CI −0.005, +0.006) for a Cox extension and −0.009 (−0.017, −0.002) for a boosted extension, which ranks slightly below the score alone. The established five-variable score is effectively at the information ceiling of this cohort.

4.5 Generalisation: complexity helps only as data grow

Replicating the comparison on three external clinical datasets (Table 3; Supplementary Fig. S2) indicated that the limited value of complexity is consistent with a sample-size-dependent pattern and is not specific to the cirrhosis cohort. These datasets are single-event, so they bear only on the complexity-versus-Cox question, and not on the competing-risks or clinical-score comparisons central to the cirrhosis analysis. On the small-to-moderate datasets, complex models barely exceeded Cox (METABRIC, $n = 1904$: best gain +0.008; GBSG, $n = 2232$: +0.014), and DeepSurv fell below Cox on METABRIC only (−0.022). Only on the large SUPPORT dataset ($n = 8873$) did they gain materially (up to +0.046). The cirrhosis cohort ($n = 418$) lies at the small-data end of this range, where little benefit from complexity should be expected, consistent with no model exceeding the clinical score.

4.6 Both competing endpoints are well predicted

Under cross-validation, both endpoints were discriminated well (Table 4). The death endpoint reached Harrell's C = 0.826 (95% CI 0.816, 0.836). The *transplant* endpoint reached Harrell's C = 0.822 (95% CI 0.799, 0.843) and Uno's C = 0.813 (0.790, 0.836) when modelled as a competing event, encouraging discrimination, though necessarily imprecise given only 25 events. On the same data, the classification baseline (Table 5, panel A) reached transplant-class F1 0.007 to 0.180, macro-F1 0.505 to 0.542 and accuracy 0.693 to 0.742. An F1 score and a concordance index are not comparable, and only the survival formulation yields a time-resolved absolute risk of each competing event. On the 5-year competing-risks concordance (Table 5, panel B), Fine–Gray reached 0.842

Table 4: Cross-validated discrimination for each competing endpoint (cause-specific Cox; 95% bootstrap CI).

Endpoint	Events	Harrell's C	Uno's C
Death	161	0.826 [0.816, 0.836]	0.779 [0.764, 0.793]
Transplant	25	0.822 [0.799, 0.843]	0.813 [0.790, 0.836]

Table 5: Classification baseline and competing-risks concordance under cross-validation (mean [95% bootstrap CI]). Panel A: three-class classifiers of the status at end of follow-up (50 folds), the mean on the first row of each classifier and its 95% CI on the second; RF, random forest; GB, gradient boosting. Panel B: truncated inverse-probability-of-censoring-weighted concordance for competing risks of the predicted 5-year cumulative incidence [46].

A. Classification baseline

Classifier	F1 alive	F1 transplant	F1 death	Macro-F1	Accuracy
RF	0.800	0.007	0.710	0.505	0.742
	[0.790, 0.810]	[0.000, 0.020]	[0.694, 0.725]	[0.497, 0.515]	[0.731, 0.753]
RF + SMOTE	0.774	0.178	0.676	0.542	0.709
	[0.765, 0.782]	[0.134, 0.220]	[0.660, 0.692]	[0.526, 0.560]	[0.698, 0.719]
GB	0.783	0.105	0.692	0.527	0.724
	[0.773, 0.792]	[0.064, 0.149]	[0.677, 0.708]	[0.510, 0.543]	[0.713, 0.735]
GB + SMOTE	0.758	0.180	0.673	0.537	0.693
	[0.746, 0.769]	[0.139, 0.224]	[0.657, 0.689]	[0.519, 0.555]	[0.681, 0.706]

B. Competing-risks C-index at 5 years

Model	Death	Transplant
Fine–Gray (4 predictors)	0.842 [0.831, 0.852]	0.807 [0.781, 0.832]
DeepHit, two-event (4 predictors)	0.825 [0.814, 0.835]	0.764 [0.732, 0.795]
DeepHit, two-event (full covariates)	0.831 [0.821, 0.842]	0.695 [0.658, 0.731]

(95% CI 0.831, 0.852) for death and 0.807 (0.781, 0.832) for transplant, with Harrell's C 0.826 and 0.806. Two-event DeepHit reached 0.825 (0.814, 0.835) for death with the four endpoint predictors and 0.831 (0.821, 0.842) with the full covariates, and 0.764 (0.732, 0.795) and 0.695 (0.658, 0.731) for transplant. The four-predictor Fine–Gray model has the highest competing-risks concordance for both causes, two-event DeepHit approaches it for death, and the full covariate set gives no gain for transplant, consistent with the static variables setting the ceiling.

4.7 Cause-specific predictors: opposite age effects

Cause-specific Cox models (Table 6; Supplementary Fig. S3) gave the effect of each covariate on each event. Death was driven by older age (HR 1.36 per SD, 95% CI 1.16–1.59), higher bilirubin (HR 1.89, 1.57–2.28), edema (HR 2.26, 1.32–3.87) and advanced fibrosis (stage 4 HR 1.56, 1.10–2.22), with albumin protective. For age, the transplant hazard moved in the *opposite* direction (HR 0.74 per SD, 0.58–0.95): younger patients with deteriorating liver function (higher bilirubin, HR 1.30, 1.01–1.67) were preferentially transplanted, while older patients with the same biochemistry died. The Fine–Gray subdistribution hazard ratios (Table 6) were 1.52 (1.26–1.82) for age on death and 0.32 (0.22–0.47) on transplant, and 2.02 (1.64–2.49) and 1.30 (0.91–1.87) for log bilirubin. The subdistribution HR for age on transplant lies further from 1 than the cause-specific HR because

Table 6: Cause-specific (Cox) and subdistribution (Fine–Gray) hazard ratios [95% CI], per SD for continuous covariates. Transplant models use age, log bilirubin and albumin; n.i., not in the model. Selected covariates; full models in the repository.

Covariate	Death HR	Transplant HR	Death SHR	Transplant SHR
Age	1.36 [1.16, 1.59]	0.74 [0.58, 0.95]	1.52 [1.26, 1.82]	0.32 [0.22, 0.47]
Bilirubin (log)	1.89 [1.57, 2.28]	1.30 [1.01, 1.67]	2.02 [1.64, 2.49]	1.30 [0.91, 1.87]
Albumin	0.80 [0.68, 0.95]	0.88 [0.68, 1.13]	0.84 [0.70, 1.00]	0.80 [0.48, 1.32]
Prothrombin (log)	1.20 [1.04, 1.38]	n.i.	1.29 [1.10, 1.51]	n.i.
Edema	2.26 [1.32, 3.87]	n.i.	2.32 [1.14, 4.74]	n.i.
Stage 4	1.56 [1.10, 2.22]	n.i.	1.50 [1.03, 2.16]	n.i.

older patients also die sooner, which removes them from transplant eligibility: the cause-specific HR describes the transplant rate among patients alive and untransplanted, and the subdistribution HR the cumulative incidence of transplant [33, 35]. In this cohort, age therefore acted in opposite directions on the two events, a pattern consistent with referral and listing practice. Because the cohort was enrolled in 1974–1984, before MELD-based allocation, the likely mechanism is era-appropriate candidate selection. A competing-risks analysis of the modern transplant waiting list found high wait-list mortality among patients with PBC listed for liver transplantation [21]. The pattern serves here as a validity check, which the competing-risks treatment reproduces directly. An end-of-follow-up status label scored by F1 reports neither effect, and the simulation (Section 4.8) shows which classifier formulations recover the underlying risk.

4.8 Simulation: recovery of the true transplant risk

On data simulated from the cohort (Fig. 2), the random forest's transplant probability ranked the true 5-year transplant cumulative incidence with Spearman 0.56 at $n = 418$ (2.5 to 97.5 percentiles 0.40 to 0.68), and 0.59 with SMOTE, against 0.91 for multinomial logistic regression, 0.92 for cause-specific Cox and 0.90 for Fine–Gray. Harrell's C against the truth was 0.69 and 0.71 for the two forests and 0.88, 0.89 and 0.88 for the other three. The gap persisted with more data: the random forest reached 0.55 at $n = 1000$ and 0.61 at $n = 4000$, while the other three reached 0.94 to 0.97 at $n = 1000$ and 0.97 to 0.99 at $n = 4000$. For death, the forests ranked the truth at 0.74 and 0.70 at $n = 418$, against 0.98 to 0.99 for the other three. Transplant-class F1 stayed between 0.14 and 0.25 for the three classifiers at every n, and between 0.15 and 0.17 for multinomial logistic regression, whose ranking is near the truth, so F1 does not measure recovery of transplant risk. Cause-specific Cox, Fine–Gray and multinomial logistic regression recovered the sign of the age effect on both outcomes in 100% of replicates at every n. At $n = 418$ their mean age coefficients per year of age were -0.094 (target -0.091) for transplant and 0.040 (target 0.039) for death with cause-specific Cox, -0.105 (-0.105) and 0.044 (0.046) with Fine–Gray, and -0.092 (-0.092) and 0.046 (0.043) with the multinomial model. The loss of transplant risk is therefore specific to tree-ensemble classifiers on a status label, scored by F1: a multinomial model fitted to the same label recovers the ranking.

4.9 Dynamic prediction on the longitudinal cohort

Modelling the biomarker trajectory in place of a single snapshot both reproduced and sharpened the competing-risks picture. In time-varying cause-specific Cox models (Table 7), the most recent measurements were strongly prognostic for death (bilirubin HR 1.82 per SD, albumin 0.63, prothrombin 1.40, edema 2.47; all $p < 0.001$), and the *opposite* age effect persisted in the longitudinal setting

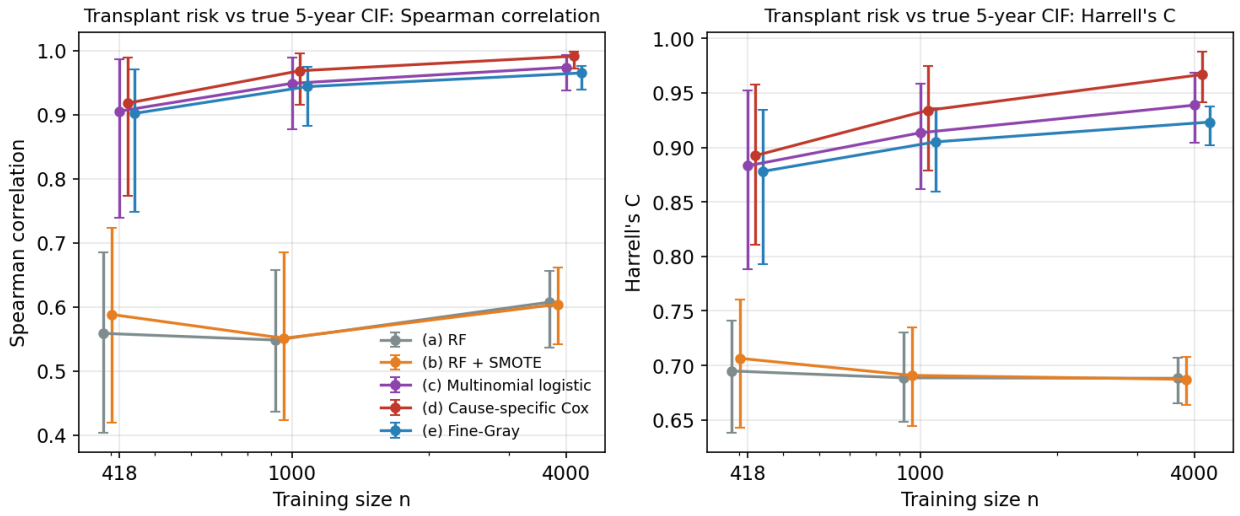


Figure 2: Simulation fitted to the cohort (cause-specific Weibull hazards; fixed test set of 5000 patients). Left: Spearman correlation between each approach's transplant risk and the true 5-year cumulative incidence. Right: Harrell's C against the truth. Points are means and bars the 2.5 to 97.5 percentiles over 200 replicates at each training size n (log scale).

Table 7: Time-varying cause-specific hazard ratios on the longitudinal cohort (per SD for continuous covariates).

Covariate	**Death HR [95% CI]**	**Transplant HR [95% CI]**
Bilirubin (log)	1.82 [1.58, 2.09]	1.28 [1.07, 1.52]
Albumin	0.63 [0.55, 0.72]	0.87 [0.73, 1.04]
Prothrombin (log)	1.40 [1.27, 1.55]	1.15 [0.97, 1.37]
Age	1.25 [1.10, 1.42]	0.83 [0.69, 0.98]
Edema	2.47 [1.72, 3.56]	1.28 [0.74, 2.20]

(death HR 1.25, $p < 0.001$; transplant HR 0.83, $p = 0.03$). The landmark analysis (Fig. 3) shows why updating matters. A baseline measurement loses prognostic value as it ages: 3-year death-risk concordance fell from 0.87 at the 2-year landmark ($n = 278$, 55 deaths) to 0.64 at 6 years ($n = 166$, 26 deaths), whereas using the most recent measurement decayed far more slowly, from 0.89 to 0.78 over the same span. The cross-validated paired advantage grew with follow-up and excluded zero at every landmark (ΔC = +0.011 [95% CI +0.004, +0.017] at 2y; +0.059 [+0.042, +0.077] at 4y ($n = 225$, 34 deaths); +0.139 [+0.116, +0.162] at 6y). The later landmarks rest on few events, and the intervals reflect cross-validation fold variability without independent resampling. The pattern suggests that, in these data, recent measurements carry information a stale baseline cannot.

4.10 Calibration and risk stratification

The Cox model produced well-calibrated five-year predictions against Kaplan–Meier estimates (Supplementary Fig. S4), and its out-of-fold predicted risk separated patients cleanly into low-, intermediate- and high-risk groups (Supplementary Fig. S5). The Fine–Gray model was assessed for both competing events against the Aalen–Johansen incidence (Supplementary Fig. S6). For death, predicted and observed 5-year cumulative incidence by quintile were 0.06 and 0.05, 0.11 and 0.03, 0.18 and 0.16, 0.34 and 0.44, and 0.78 and 0.79, so predictions follow the observed incidence

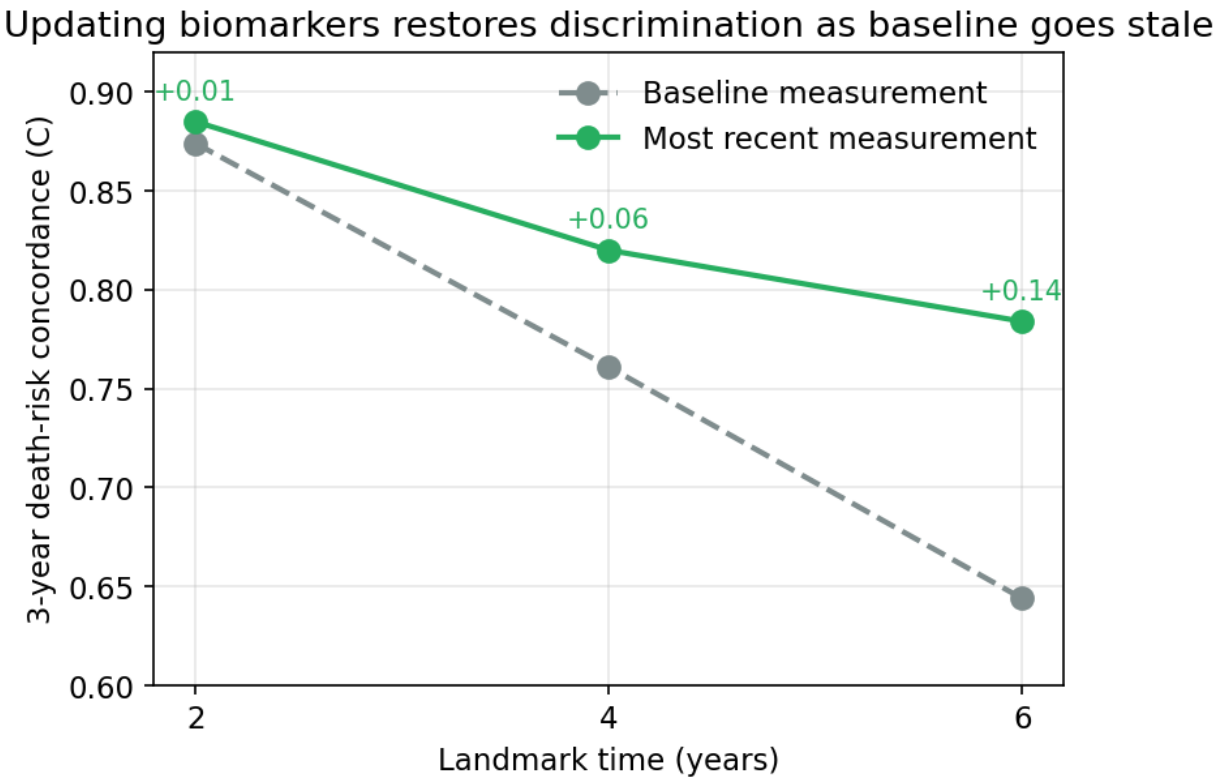


Figure 3: Landmark analysis on the longitudinal cohort. As the baseline measurement ages, its 3-year death-risk discrimination decays; the most recent measurement restores it, with a widening advantage.

across predicted risks from 0.06 to 0.78, and every predicted value lies inside the observed 95% interval except in the second quintile (observed interval 0.00 to 0.07). For transplant they were 0.005 and 0.012, 0.012 and 0.000, 0.023 and 0.014, 0.049 and 0.055, and 0.136 and 0.144. The second quintile had no transplant within five years, and the other four predicted values lie inside their observed intervals. Calibration of this kind is a prerequisite for absolute-risk communication.

4.11 Net benefit for 5-year death

In the competing-risks decision-curve analysis (Fig. 4), every model's net benefit exceeded both treat-all and treat-none across thresholds 0.05 to 0.50, with the smallest margins at 0.05 (gradient boosting 0.0013, Mayo score 0.0015, random survival forests 0.0035, Fine–Gray 0.0038, Cox PH 0.0049). The Mayo score had the highest net benefit at 34 of the 46 thresholds, Cox PH at 8 and Fine–Gray at 4. The largest absolute differences from the Mayo score over the 46 thresholds were 0.0145 for Cox PH, 0.0172 for Fine–Gray, 0.0183 for gradient boosting and 0.0320 for random survival forests, and Cox PH, Fine–Gray, random survival forests and gradient boosting exceeded the Mayo score at some thresholds by at most 0.004, 0.004, 0.003 and 0.001. At a threshold of 0.20, net benefit was 0.214 (95% CI 0.164, 0.260) for the Mayo score, 0.206 (0.159, 0.251) for Cox PH, 0.206 (0.159, 0.251) for Fine–Gray, 0.197 (0.149, 0.242) for gradient boosting and 0.183 (0.131, 0.235) for random survival forests, against 0.115 (0.055, 0.171) for treat-all. Treat-all net benefit was positive up to a threshold of 0.29. Cox PH and Fine–Gray therefore match the Mayo score's

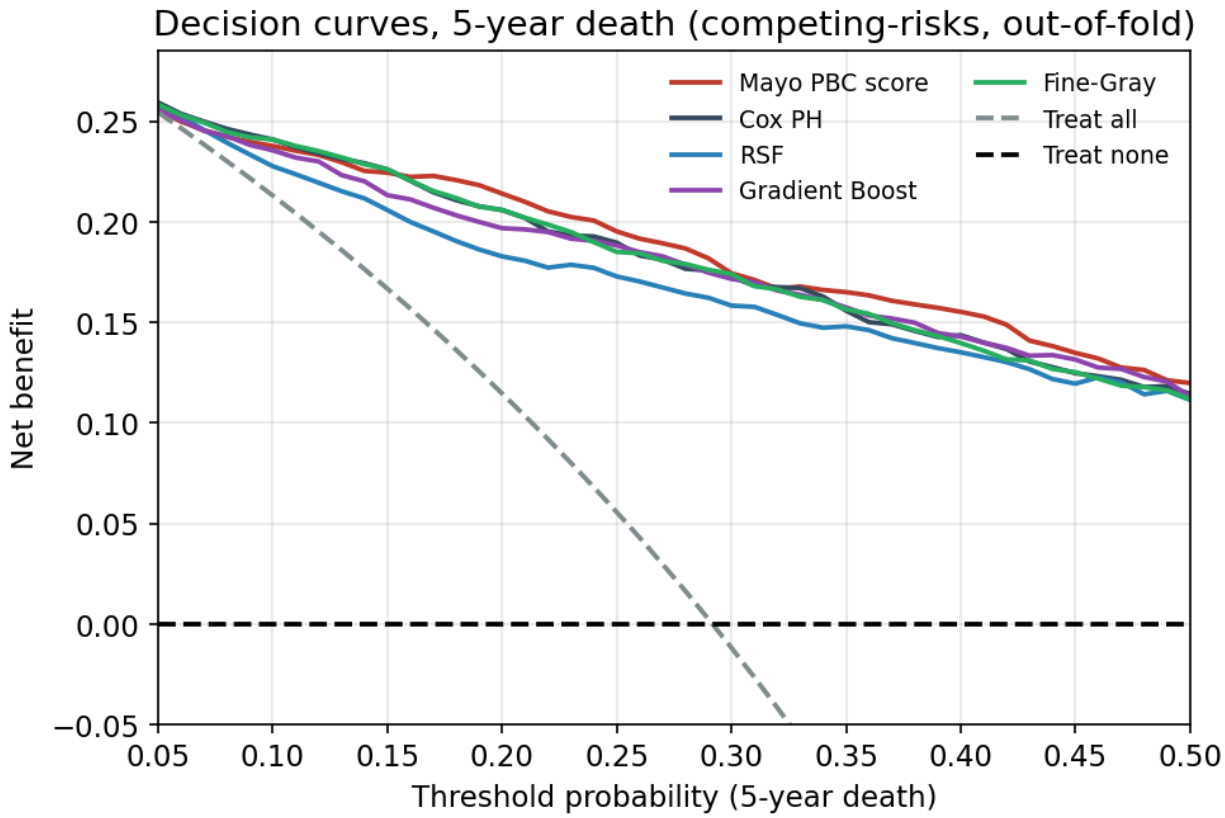


Figure 4: Competing-risks decision curves for death at 5 years: Mayo PBC score, Cox PH, random survival forests (RSF), gradient boosting, Fine–Gray, treat all and treat none, over thresholds 0.05 to 0.50. Every risk is the out-of-fold predicted 5-year cumulative incidence of death with transplant as a competing event (4 cross-validation repeats), and net benefit uses the Aalen–Johansen incidence among patients above the threshold.

net benefit to within 0.02 at every threshold, consistent with the ceiling set by the static variables.

4.12 Transportability to an independent cirrhosis cohort

Discrimination did *not* transport to the SUPPORT cirrhosis subgroup ($n = 508$, 329 deaths). The 3-marker PBC model, which reached internal cross-validated C = 0.823, fell to C = 0.558 externally, barely above chance, and refitting the same markers within SUPPORT reached C = 0.594. The albumin coefficient reversed sign, from protective in PBC (HR 0.38) to mildly harmful in SUPPORT (HR 1.16). This is best read as population specificity. SUPPORT enrolled acutely hospitalised patients (65% died; median follow-up 234 days), a fundamentally different phenotype from the chronic outpatient PBC trial. Chronic-PBC prognostic structure therefore does not carry to acute decompensated cirrhosis, a caution that prognostic models, like the formulation itself, are tied to clinical context. A fair external test would require an independent *chronic* cirrhosis cohort with the full marker set, which is not currently available in the public domain.

5 Discussion

The main result comes from the simulation. On data generated from the cohort, tree-ensemble classifiers of the status label scored by F1, which we test as representative of the classification framing, recover the true transplant-risk ranking poorly (Spearman 0.55 to 0.61), cause-specific Cox, Fine–Gray and multinomial logistic regression recover it (0.90 to 0.99), and transplant F1 (0.14 to 0.25) does not separate the two. The risk is lost through the classifier family and the scoring rule, since a multinomial model on the same status label keeps it. On the real data, transplant modelled as a competing event reaches a cross-validated concordance of 0.82 (25 events; 95% CI 0.80–0.84), an encouraging if imprecise result. The cause-specific and subdistribution models give opposite age effects (cause-specific HR 1.36 and 0.74; subdistribution HR 1.52 and 0.32), in which younger patients with failing livers are transplanted while older ones die, consistent with referral and listing practice in PBC, where wait-list mortality among listed patients is high [21]. We present this as a validity check: the competing-risks formulation reproduces directly a pattern expected from listing practice.

The seven-model benchmark is the evidence that the static variables set the ceiling, and that the formulation and updated measurements are where the gains lie. Across seven model families spanning a five-variable clinical score, penalized linear models, tree ensembles and deep neural survival networks, discrimination was flat at C $\approx$ 0.83, and no model exceeded the decades-old Mayo score. Augmenting the score with additional biomarkers did not raise it, and paired per-fold tests showed that no model significantly exceeded the score: Cox and penalized Cox were indistinguishable from it, and random survival forests, gradient boosting, DeepSurv and DeepHit ranked slightly below it. The ceiling is therefore limited by the information in this static set of variables. Refitting the five Mayo variables within each fold matched the externally-derived score (C 0.832 vs. 0.833; paired ΔC -0.001 [-0.006, $+0.004$]), ruling out an in-sample-baseline artefact, and the longitudinal analysis shows that the static snapshot is the binding constraint on the achievable signal. The competing-risks models extend the ceiling: Fine–Gray reached a 5-year competing-risks concordance of 0.842 for death and 0.807 for transplant, and two-event DeepHit 0.825 and 0.831 for death and 0.764 and 0.695 for transplant, with the four endpoint predictors and the full covariates. In the decision-curve analysis, Cox PH and Fine–Gray matched the Mayo score's net benefit for 5-year death (largest differences 0.0145 and 0.0172 over thresholds 0.05 to 0.50), each exceeding it at some thresholds. Our replication across three external (single-event) datasets is consistent with the complexity benefit being size-dependent. Complex models exceeded Cox only marginally on cohorts of about two thousand patients and meaningfully only on the largest ($n \approx 9000$). These datasets speak only to the complexity-versus-Cox question, but they place $n = 418$ where little benefit should be expected. This replicates, in the survival setting, an established pattern on general tabular data, where simple and tree-based models are difficult to beat with deep networks at limited sample sizes [30, 31]. Competing-risks models have been benchmarked on these data before [19, 20], by Sierra et al. with six models (cause-specific Cox, Fine–Gray, penalised Fine–Gray, random survival forests, DeSurv and DeepHit), a set that includes ours, and decision-curve analysis at 5 years [20]. Neither study compares the models with the Mayo score. Obite et al. [25] compared survival models on this cohort by concordance and Brier score, reporting that a factor-enhanced DeepSurv outperformed several existing survival models. The present study adds the Mayo comparator, including the Mayo score as a decision-curve strategy, and the simulation.

The practical implication is that the value of survival ML here lies in its *formulation and outputs*. A competing-risks survival model delivers calibrated, time-resolved survival estimates and cause-specific risk for both death and transplantation, handles censoring and the competing event, and requires no manual scoring. For 5-year death, the competing-risks models reach the Mayo score's

net benefit (largest differences 0.0145 for Cox PH and 0.0172 for Fine–Gray). These are outputs a classifier or a fixed score does not provide, and they come at no cost in discrimination.

A further result points forward. On the longitudinal cohort, the prognostic value of a measurement decayed markedly with time, with baseline death-risk concordance falling to 0.64 by six years, but was largely recovered by using the most recent measurement (0.78), the dynamic advantage widening throughout follow-up. This single-cohort, landmarking-based result is a proof of concept rather than a validated monitoring strategy. Still, it suggests that the prognostic value in these data may lie less in any single snapshot than in *serial monitoring*, motivating competing-risks prediction from updated laboratory values, which would extend the time-varying survival models developed for cirrhosis [24] to competing events.

Limitations. The cohort is single-centre, historical and modest ($n = 418$), and the transplant endpoint has only 25 events, roughly eight per covariate in the inferential model, so its estimates, though encouraging, are wide and at risk of optimism. Our primary analysis uses a single snapshot per patient. Although we analysed the longitudinal version of the cohort for the dynamic analyses, we used landmarking in place of full joint longitudinal-survival modelling, which may extract further signal. Our attempt at external validation was limited by data availability. The only freely-available independent cirrhosis cohort (SUPPORT) is an acutely hospitalised population too dissimilar from the chronic outpatient PBC trial for a fair test, and the model did not transport to it. A fair external validation awaits an independent chronic cirrhosis cohort with the full marker set. The data are also PBC-specific, so generalisation to other cirrhosis aetiologies is unverified. The simulation mechanism has proportional cause-specific hazards that are linear in five covariates, the form of the cause-specific Cox model. The multinomial logistic model, which is not a hazard model, recovers the ranking as well, so the forests' gap does not follow from the hazard form alone, but mechanisms with interactions or non-proportional hazards were not simulated. The proportional-hazards assumption held for most covariates but was violated for prothrombin time (Schoenfeld $p = 0.002$); because the analysis is discrimination-focused this has limited impact, but it qualifies the cause-specific hazard ratio for that covariate. Finally, MELD could not be benchmarked because its inputs are absent from the dataset.

Future work. Natural extensions are external validation on contemporary multi-centre cohorts and joint longitudinal-survival modelling that goes beyond the landmarking used here.

6 Conclusion

A widely-used cirrhosis ML benchmark is often analysed as classification of its end-of-follow-up status label, a framing that answers a weaker question than prognosis requires. In a simulation fitted to the cohort, forest classifiers ranked the true transplant risk at Spearman 0.55 to 0.61, against 0.90 to 0.99 for cause-specific Cox, Fine–Gray and multinomial logistic regression, while transplant F1 stayed at 0.14 to 0.25 for every classifier. Recast as competing-risks survival, the transplant label becomes an interpretable competing-risks outcome with encouraging, if imprecise, discrimination (C = 0.82; 25 events), and the analysis recovers the opposite-direction age effects on death and transplantation found in this cohort. The seven-model benchmark supports this step. No survival model, including deep networks, exceeds the simple, validated Mayo clinical score, which places the ceiling in the available static variables. Refitting the score within folds reproduced it (C 0.832 vs. 0.833), ruling out an in-sample artefact, and the size-dependence of the benefit of complexity is consistent with the external datasets. For 5-year death, Cox PH and Fine–Gray match the Mayo score's net benefit (largest differences 0.0145 and 0.0172 over thresholds 0.05 to 0.50). Longitudinally, updating biomarkers recovers discrimination that a stale baseline loses,

suggesting value in serial monitoring. On this benchmark, and consistent with broader tabular-ML evidence, the problem formulation, dynamic prediction and calibrated, cause-specific outputs matter more than model complexity. The lesson transfers beyond cirrhosis: censored benchmarks with competing events should be scored on the time-resolved absolute risk of each event with competing-risks methods, in place of status-label F1.

Data availability

All datasets are publicly available: the primary cohort from the UCI Machine Learning Repository ("Cirrhosis Patient Survival Prediction"), the longitudinal version (`pbcseq`) from the R `survival` package, and the external datasets via `pycox` and the SUPPORT study. The complete, leakage-free analysis pipeline, covering the competing-risks formulation, the classification baseline, the seven-model benchmark, the Fine–Gray and two-event DeepHit models, the decision-curve analysis, the simulation study, dynamic landmarking, evaluation and figures, is released as a reproducible framework intended for reuse on this and comparable clinical cohorts. It is available at `https://github.com/TheAxeC/cirrhosis-competing-risks`, and will be archived with a citable Zenodo DOI on acceptance.

Ethics

This study used fully anonymised, publicly available datasets and did not involve new human-subjects research. No ethics approval was required.

CRediT author contributions

Meseret Assefa Kerga: conceptualization, methodology, software, formal analysis, investigation, writing – original draft. **Axel Faes:** conceptualization, methodology, supervision, writing – review & editing.

Declaration of competing interest

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During the preparation of this work the authors used a generative AI assistant (Claude) to support software implementation and to edit and improve the language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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